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Midterm Examination, Business 33801:

(Monday, 11 December 2023)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 9 questions with a total of 90 points possible. You have 90 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

I pledge my honor that I have not violated the Honor Code during this examination.

Sign your name here.

True-False-Uncertain (30 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and EXPLAIN your answer. Read each question carefully. You may take the non-italicized portion of the statement as true. **No credit will be given for answers without explanation.**

Question 1 (6 points) True, False, Uncertain? [Explain your answer.] *A price-taking firm with a U-shaped average cost curve always produces in the region with economies of scale if it operates.*

Question 2 (6 points) True, False, Uncertain? [Explain your answer.] Barbie spends all of her disposable income on two goods: fashion accessories, a , and beach umbrella rentals, b . Both are normal goods. The prices are $p_a = 4$ per fashion accessory (they are all priced the same), and $p_b = 2$ per hour of beach umbrella rental. Each month, she purchases 10 accessories and 30 hours of beach umbrella rentals. This month, the local beach umbrella rental company is celebrating their 10-year anniversary and, in appreciation of Barbie's loyalty, is giving her 10 free hours of beach umbrella rental. The prices of accessories and beach umbrella rentals remain as before. *Barbie will buy greater than 10 accessories this month.*

[Hint: sketch her budget constraint before and after if you are unsure about what happens.]

Question 3 (6 points) True, False, Uncertain? [Explain your answer.] The market for nickel has a constant demand elasticity of $\varepsilon^d = -\frac{1}{2}$ and a supply elasticity of $\varepsilon^s = 2$. Following an armed conflict, several mines shut down and the supply curve for nickel shifts leftward with 10 percent less nickel supplied at every price than before. *Market expenditures on nickel increase.*

Question 4 (6 points) True, False, Uncertain? [Explain your answer.] Skipper runs a consulting firm that produces economic reports for her clients. To produce reports, she can use any combination of highly skilled University of Chicago graduates (skilled labor, x) and undergraduate students who have only taken a few economics classes (unskilled labor, y). The production function for generating reports is $f(x, y) = 0.25x + 0.10y$. Skilled labor costs \$200 per hour; unskilled labor costs \$50 per hour. *Skipper should only hire unskilled labor in her consulting firm.*

Question 5 (6 points) True, False, Uncertain? [Explain your answer.] Greta runs a profit-maximizing, price-taking firm. *If she decides to produce, she will choose a level of output where the average opportunity cost curve is at its minimum.*

Short problems (60 points total)

Question 6 (18 points) Consider a competitive market with a demand function given by

$$Q^d = 12,000 - 300p.$$

This demand relationship holds for both the short run and the long run.

In the short-run, there are presently 200 firms competing, each with a short-run opportunity cost function given by

$$C_{sr}^o(q) = q^2.$$

In the long run, every firm's opportunity cost function is

$$C_{lr}^o(q) = 100 + q^2,$$

and the assumptions of a constant-cost industry are satisfied.

(a). What is the short-run equilibrium price and quantity in the market? [You may assume that all 200 firms find it optimal to produce.]

(b). What is the long-run equilibrium price and quantity in this market?

Question 7 (14 points) Ken has an income of $I = 10,000$ each year which he spends on only two things – roller skates, s , and fur coats, c . The price of a pair of roller skates is $p_s = 100$ and the price of a fur coat is $p_c = 250$. He currently buys $s^* = 50$ pairs of roller skates and $c^* = 20$ fur coats each year. Assume that Ken's optimal consumption point is "regular" (i.e., his indifference curves are smooth and convex and the optimal point satisfies the textbook tangency condition).

(a). Draw the budget line (placing fur coats on the horizontal axis), illustrate the axes, the intercepts and the slope of the budget line. Also draw an indifference curve for Ken that is consistent with his choice of $s^* = 50$ and $c^* = 20$.

(b). Suppose that the price of roller skates decreases to $p_s = 80$ and the price of coats increases to $p_c = 300$. (Ken's income remains unchanged at $I = 10,000$.) At these new prices, Ken chooses to consume more roller skates and fewer fur coats. After the prices change, is Ken better off, worse off, or is it not possible to say. Explain. [Hint: check how much Ken's original consumption bundle would have cost under the new prices.]

Question 8 (18 points) Mojo Dojo Casa Corporation (MDC) is a price-taking firm that sells flip-flop sandals. It can sell as many pairs of flip-flops as it pleases at a price of $p = 100$ per pair. Its total opportunity cost of producing q pairs of flip flops is

$$C^O(q) = 40q + q^2.$$

[Hint: the derivative of this function is $40 + 2q$.]

(a). How many pairs of flip flops (if any) should MDC produce and sell, and what (if any) "economic" profit does it earn?

(b). Before making the production decision this period, MDC discovers that for an additional fixed set-up cost of \$100, it can produce a higher-quality flip flop using the same production facility. The production cost for $q > 0$ pairs of high quality flip flops is therefore $100 + 40q + q^2$. The high-quality flip flops, however, can be sold for $p = 200$ per pair. Unfortunately, MDC can only use its factory to produce one type of flip flop; it can produce regular flip flops or high-quality flip flops, but *MDC cannot make both types*.

What is the opportunity cost of producing q pairs of high-quality flip flops at the factory?

(c). Keeping the facts of part (b) above, what is the profit-maximizing production decision for MDC (i.e., which type of flip flop and how many (if any) should be produced)?

Question 9 (10 points) Midge (who has no wealth in the bank) is offered a choice between (1) a payment of \$10,000 and (2) a 60 percent chance of \$40,000 and a 40 percent chance of nothing.

(a). If Midge's utility function for money is $u(x) = \sqrt{x}$, which should she choose – the certain payment or the risky outcome? Explain.

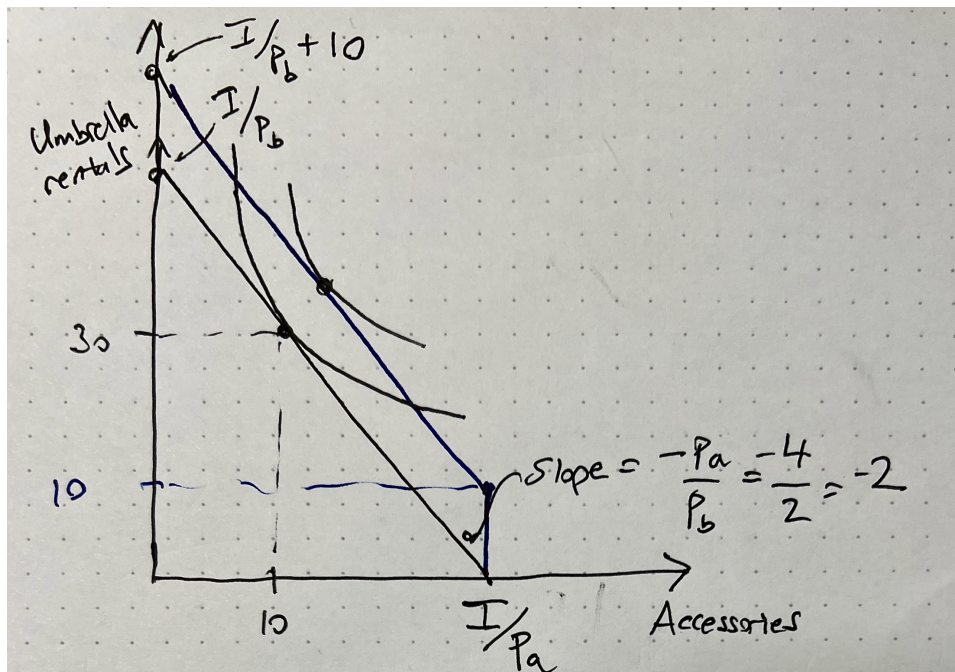
(b). Suppose that Midge has the risky portfolio in (a) – a 60 percent chance of \$40,000 and a 40 percent chance of nothing. What is the risk premium she associates with this portfolio?

Suggested answers to examination

1 False. We know that an operating firm chooses q^* such that $p = MC(q^*)$ and that $p \geq AC^O(q^*)$. Thus, $MC(q^*) = p \geq AC^O(q^*)$. But $MC(q^*) \geq AC^O(q^*)$ implies that average opportunity cost is nondecreasing which is inconsistent with economics of scale (i.e., economies of scale would exhibit $MC(a^*) < AC^O(a^*)$).

2 True. Because the prices are unchanged, getting the first 10 hours of rental for free is equivalent to Barbie being given extra income in the relevant region of her budget constraint (i.e., $\Delta I = 10p_b = 20$). Because fashion accessories are a normal good, she will consume more accessories this month than before.

Aside: This is clear if you look at the budget constraints both before and after the free rentals: the new budget line is a parallel outward shift of the original one. Because both goods are normal, she will consume more of each (i.e., the new optimal choice point must lie to the north-east of the original choice).



3 True. Because the supply curve shifts left, the original and new equilibrium points lie on the unchanging demand curve. Because the demand curve is inelastic, a reduction in output (i.e., a movement up along the demand curve) leads to an increase in market expenditures (i.e., a larger revenue rectangle).

4 True. Skipper will only want to hire unskilled labor. There are at least two ways to explain the answer.

The marginal rate of technical substitution between skilled and unskilled labor is

$$MRTS_{x,y} = \frac{MP_x}{MP_y} = \frac{.25}{.10} = 2.5.$$

The input price ratio, however, is

$$\frac{w_x}{w_y} = \frac{200}{50} = 4 > 2.5.$$

Thus, Skipper values skilled labor (relative to unskilled labor) less than the market values skilled labor (relative to unskilled labor). Hence, hire more unskilled labor and less skilled labor. Because the marginal products are constant, this implies only hiring unskilled labor.

Alternatively, consider the reports generated per dollar spent on each for of labor. One dollar spent on skilled labor generates

$$\frac{MP_x}{w_x} = \frac{0.25}{200} = .00125$$

reports per dollar spent on skilled labor. Alternatively, we have only

$$\frac{MP_y}{w_y} = \frac{0.10}{50} = .0020$$

reports generated per dollar spent on unskilled labor. Unskilled labor has a higher rate of return than skilled labor. Hence, Skipper should hire only unskilled labor.

5 False/Uncertain. Greta should produce at a point where marginal cost equals marginal revenue (the marginal output rule). There is no reason *a priori* why this should arise at the point where average cost is at a minimum.

6 (a). Each firm's marginal output rule requires that $p = MC_{sr}^o(q)$, and so $p = 2q$. Consequently, each firm's supply function is

$$q^s = \frac{1}{2}p.$$

The short-run market supply is therefore 200 times this,

$$Q_{sr}^s = 200q^s = 100p.$$

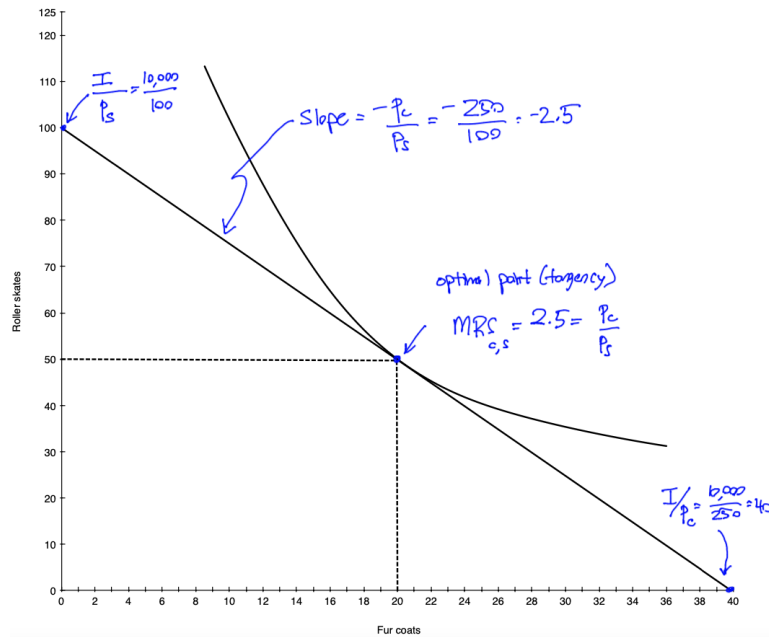
Equating supply with demand, we have the following equilibrium condition:

$$100p = 12000 - 300p$$

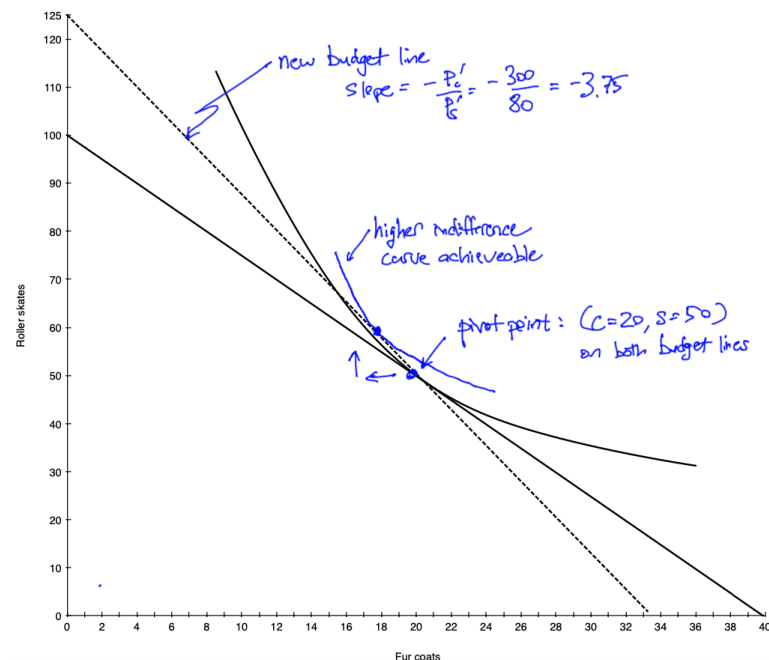
and thus $p^* = 30$. At this price, $Q^* = 100(30) = 3000$.

(b). In this constant-cost industry with free entry, the industry supply curve will be a horizontal line (i.e., perfectly elastic). In order to find the price at which supply is perfectly elastic, we need to identify the minimum of the average cost curve, which is where average cost equals marginal cost. From the total cost function, we have that $AC^o(q) = \frac{100}{q} + q$ and $MC(q) = 2q$. At their intersection, $\hat{q} = 10$. The corresponding average cost at $\hat{q}$ is 20. Therefore, the supply curve will be perfectly elastic at $p = 20$, which implies that the equilibrium price is $p^* = 20$. At $p^* = 20$, the equilibrium quantity is $Q^* = 12,000 - 300(20) = 6000$.

7 (a).



(b). Note that under the new prices, Ken can still afford to purchase the bundle he used to purchase because $50(80) + 20(300) = 10,000$. Because he could buy what he had before, he can't be any worse off. Moreover, because his previous $MRS_{c,s} = \frac{250}{100} = 2.5$ is no longer equal to the new price ratio, $\frac{300}{80} = 3.75$, he can do better by choosing a new optimal bundle with more skates and fewer coats, which he does. Thus, Ken must be better off. This is shown in the following figure (though a figure was unnecessary for full marks on this part if the explanation was clear).



8 (a). The marginal output rules requires that, if the firm produces, the firm produces where $MC = MR$. In the present case, $MC = 40 + 2q$ and $MR = 100$. Thus,

$$40 + 2q = 100,$$

so 30 pairs should be produced if the firm operates. We next ask whether the firm should operate. If the firm produces 30 pairs, it will earn revenues of $R = 100(30) = 3000$ and it will have opportunity costs of $C^0(30) = 40(30) + (30)^2 = 1200 + 900 = 2100$. Thus, economic profit is

$$\pi = 3000 - 2100 = 900 > 0,$$

and we conclude that the firm should produce $q^* = 30$.

(b). The opportunity cost of producing q units of high-quality alloy is the sum of the direct production costs, $100 + 40q + q^2$, and the opportunity cost of using the factory to produce high-quality flip flops (and hence the lost profit from making standard flip flops) ($\pi = 900$ from part (a) above). Thus,

$$C_{hq}^O(q) = (100 + 40q + q^2) + 900 = 1000 + 40q + q^2.$$

(c). Using the marginal-output rule for high-quality flipflops, we have

$$MC = MR \iff 40 + 2q = 200,$$

so the firm should produce $q_{hq} = 80$ units of high-quality flip flops, if it produces at all. Checking the shut-down/operate rule, we verify that economic profit is

$$\pi_{hq} = 200(80) - 1000 - 40(80) - (80)^2 = 16,000 - 1000 - 3200 - 6400 = 16000 - 10,600 = 5400 > 0.$$

Thus the firm should produce the high-quality flip flops and instead of the low quality ones and should manufacturer $q^* = 80$ pairs of these flip flops.

9 (a). Midge's EU from the 10,000 payment is $EU = \sqrt{10,000} = 100$. Midge's EU from the gamble is

$$EU = 0.6\sqrt{40,000} + 0.4\sqrt{0} = 0.6(200) = 120.$$

Hence, Midge should take the risky gamble.

(b). Midge's certainty equivalent of the risky payment satisfies

$$\sqrt{CE} = 120,$$

so $CE = (120)^2 = 14,400$. The expected payoff of the gamble is $E[\tilde{x}] = 0.6(40,000) + 0.4(0) = 24,000$. The difference is

$$\rho = 24,000 - 14,400 = 9600,$$

which is Midge's risk premium.